Week 4 Homework

Problem 1.

a) The attention mechanism acts as a filter that allows the model to focus on the relevant parts of the input sequence. Not all tokens contribute equally to a given output position, b.c in a real-world sentence, different parts of a sentence have different relationship to the one of the other part, some parts are more closely related, some parts are not. For instance, the verb might be more closely related to a subject, than the adjective. The attention mechanism will create a weight for each pair of token in a sentence, using the dot product of query and key vector. The weight will reflect the relevance of query and key vector. The greater the relevance, the larger the weight.

b) The query is asking for which part of the sentence is most relevant to the current output. Projecting the input into a dedicated query space will allow the sentence to learn which specific token in the sentence are useful for seeking information, rather than remaining to be a raw static information

c) The key represents an answer to the query, describing what kind of information it offers. The dot product represents the closeness of the two vector, is the angle between two vector is 90° degree, it will have dot product of zero and no relevance. If they are close to each other, than it will have a large dot product.

d) The value represents the actual meaning that is passed forward. Using a seperate projection for the value is useful because it decouples the processing of finding relevant information (key) from the process of extracting the useful content, (value).

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e) They would provide different weights, b.c. the two "attentions" have different meanings, and they are therefore related to different parts of the sentence.

Problem 2:

a) $S = K^T Q$

$$S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 2}$$

(i,j) entry represents the relevence between ith key and j-th query.

b) $\sqrt{d_k} = \sqrt{2}$

$$\hat{S} = \frac{S}{\sqrt{d_k}}$$

$$= \frac{\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}}{\sqrt{2}}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

c) $\text{softmax}(\hat{S}) = A$

$$a_{11} = \frac{e^{\frac{1}{\sqrt{2}}}}{e^{\frac{1}{\sqrt{2}}} + e^{\frac{1}{\sqrt{2}}}} = 0.5$$

$a_{21} = 0.5$

$$a_{12} = \frac{e^0}{e^0 + e^{\frac{1}{\sqrt{2}}}} = \frac{1}{1+e^{\frac{1}{\sqrt{2}}}} = 0.33$$

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$a_{22} = 1 - 0.33 = 0.67$

$$A = \begin{pmatrix} 0.5 & 0.33 \\ 0.5 & 0.67 \end{pmatrix}$$

d) $Z = VA$

$$= \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0.5 & 0.33 \\ 0.5 & 0.67 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0.66 \\ 1.5 & 2.01 \end{pmatrix}$$

e) query 1 attend to both token equally

Problem 3:

a) $$\text{softmax}(S) = \begin{pmatrix} \frac{e^2}{e^2+e+1} \\ \frac{e}{e^2+e+1} \\ \frac{1}{e^2+e+1} \end{pmatrix} = \begin{pmatrix} 0.67 \\ 0.24 \\ 0.09 \end{pmatrix}$$ $0.67 + 0.24 + 0.09 = 1$

b) $$\text{softmax}(S') = \begin{pmatrix} \frac{e^4}{e^4+e^2+1} \\ \frac{e^4}{e^4+e^2+1} \\ \frac{e^4}{e^4+e^2+1} \end{pmatrix} = \begin{pmatrix} 0.87 \\ 0.12 \\ 0.01 \end{pmatrix}$$

$0.87 + 0.12 + 0.01 = 1$

c) The large raw attention score will make the weight vector really sparse, and it will lead to vanishing gradient

Problem4

a) $X \in \mathbb{R}^{512 \times 6}$ $Q \in \mathbb{R}^{64 \times 6}$

$W^Q \in \mathbb{R}^{512 \times 64}$

b) $S \in \mathbb{R}^{6 \times 6}$

c) $A \in \mathbb{R}^{6 \times 6}$

d) $Z \in 64 \times 6$

The first dimension must equal to dv because z is the matrix product of VA,

Problem 5

a) W^Q have $256 \times 64 = 16384$ parameters

W^K have 16384 parameters

W^V have 16384 parameters

A single head has 49152 parameters

Total 196608 parameters across 4 heads

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b) W^o has $4 \times 64 \times 256 = 65536$ parameters

$196608 + 65536 = 262144$ parameters in total

c) $262144 \times 6 = 1572864$ parameters in 6 layers

d) The total count changes by a factor of 4.

Problem 6:

a) Multiheaded attention allows the transformer model to simultaneously focus on different relationship within the text. Multiple heads do not have to average by different dependencies and fit them into one distribution, thus, it prevents information loss

b)

i) This suggests that the model is using its capacity redundantly. It suggests that there might be too much parameter for the task.

ii) Dropping heads may remove redundant parameters, it can also increase the

diversity of the information learned by randomly zeroing certain heads, allowing transformer to learn information from different set of heads.

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